MIL-H-3407A(CmlC)
5 June 1958
SUPERSEDING
MIL-T-3407
15 February 1951

MILITARY SPECIFICATION
HOISTING UNIT, TRIPOD, DRUM, M1

1. SCOPE

1.1 This specification covers a hoisting unit for use in lifting 55 gallon containers during filling operations in the field.

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issue in effect on date of invitation for bids, form a part of this specification:

SPECIFICATIONS

FEDERAL

PPP-T-60	- Tape; Pressure Sensitive Adhesive, Waterproof - For Packaging and Sealing.
PPP-B-621	- Boxes, Wood, Nailed and Lock-Corner
PPP-B-636	- Boxes, Fiber.

MILITARY

MIL-B-121	- Barrier Material, Greaseproofed, Flexible, (Waterproofed)
MIL-L-3150	- Lubricating - Oil, Preservative, Medium

STANDARDS

MILITARY

MIL-STD-10	- Surface Roughness, Waviness, and Lay
MIL-STD-105	- Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-129	- Marking for Shipment and Storage

FSC 3950

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DRAWINGS

CHEMICAL CORPS

B28-12-4 - Hoisting Unit, Tripod, Drum, Ml - Bill of Material.

(Copies of specifications. standards, drawangs, and publications required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 Materials and components.

3.1.1 Materials. - All materials cited on Drawing B28-12-4 or on the subsidiary drawings, shall conform to the specifications thereon, or the specific characteristics set forth on the drawings.

3.1.2 Components. - All components of the item shall conform to the specifications and drawings listed on Drawing B28-12-4 and subsidiary drawings.

3.2 Manufactured and assembly. - The hoisting unit shall be manufactured and assembled as specified on Drawing C28-12-2. Surface roughness, waviness, and lay specified by symbols on detail drawings shall comply with the applicable classes specified in Standard MIL-STD-10.

3.3 Performance. - The hoisting unit shall not require a pull of more than 125 pounds on the hand chain while causing a 1000 pound load to lift when tested as specified on 4.3. Also the hoist shall hold the load in any position without any downward creep when there is no pull on the hand chain.

4. QUALITY ASSURANCE PROVISIONS

4.1 Supplier's responsibility. - Unless otherwise specified herein the supplier is responsible for the examination and testing of the hoisting units prior to submission for acceptance.

4.2 Inspection provisions.

4.2.1 Materials and components. - The supplier shall present objective evidence as required by the Government representative that all materials and components, are in accordance with 3.1.1 and 3.1.2.

4.2.2 Lotting. -A regular production lot shall consist of the hoisting units produced by one manufacturer, from the same materials, and under essentially the same manufacturing conditions.

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4.2.3 Sampling for nondestructive inspection. - Sampling shall be conducted in accordance with Standard MIL-STD-105.

4.2.4 Examination, nondestructive. - The hoisting units shall be inspected in accordance with the classification of defects and Standard MIL-STD-105.

4.2.5 Preparation for delivery. - The preparation for delivery shall be inspected for compliance with section 5.

4.2.6 Classification of defects.

4.2.6.1 Hoisting unit, tripod, drum, M1 - assembly. - (Dwg. C28-12-2)

Categories and defects

Major — AQL 1.0 percent defective

1. Component missing
2. Leg assembly incorrect
3. Shackle not assembled to lug
4. Damage or deformation

Minor — AQL 4.0 percent defective

51. Eyebolt not peened
52. Nameplate missing, incorrect or improperly assembled
53. Coating missing or improper

Incidental — AQL 10 percent defective

101. Contamination

4.2.6.2 Fork. - (Dwg. A28-12-5)

Categories and defects

Major — AQL 1.0 percent defective

1. Thickness (min) incorrect
2. Center hole diameter (min.) incorrect
3. Center hole location incorrect
4. Slots depth (min.) incorrect
5. Slots width (min.) incorrect
6. Hole through slot (min. and max.) diameter incorrect
7. Hole through slot (min.) diameter incorrect
8. Fork wall thickness (min.) incorrect
9. Cracks and damage

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Minor; AQL 4.0 percent defective

51. Thickness (Max.) incorrect
52. Radii missing
53. Contour of fork improper
54. Rough finish

Incidental

None

4.2.6.3 Hoist, differential, 1/2-ton, steel frame.

Categories and defects	Acceptance standards	
Major A	AQL 0.25 percent defective	
1a. Function test		4.3
Major	AQL 1.0 percent defective	
1. Difficult to function		
2. Component missing		
33 Type incorrect		
4. Pull chain incorrect length		
Minor	AQL 4.0 percent defective	
51. Component damaged		
52. Rust		

Incidental

None

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4.2.6.4 Eyebolt. - (Dwg. A28-12-9)

Categories and defects

Major — AQL 1.0 percent defective

1. Eye diameter (min.) incorrect
2. Ring outside diameter incorrect
3. Thread pitch diameter and major diameter incorrect
4. Shank diameter (max.) incorrect
5. Cracks or deformation

Minor — AQL 4.0 percent defective

51. Shank length (min.) incorrect
52. Thread length (min.) incorrect
53. Rusted or contaminated threads

Incidental

None

4.2.6.5 Plate, name. - (Dwg. A22-45-2)

Categories and defects

Major — AQL 1.0 percent defective

1. Lettering incorrect
2. Material incorrect
3. Holes not drilled or punched
4. Improperly etched

Minor — AQL 4.0 percent defective

51. Size incorrect
52. Gauge incorrect
53. Holes not properly located

Incidental

None

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4.2.6.6 Sling, hoist - assembly. - (Dwg. E28-12-12)

Categories and defects

Major A — AQL 0.25 per cent defective
1a. Load Test

Major — AQL 1.0 percent defective
1. Hook incorrect or damaged
2. Inside hcok to top coupling link length incorrect
3. Pear-shaped link inside width at top incorrect

Minor — AQL 4.0 percent defective
51. Chain incorrect
52. Sharp edges

Incidental
None

4.2.6.7 Chain, leg - assembly. - (Dwg. E28-12-13)

Categories and defects

Major — AQl 1.0 percent defective
1 Length including shackle incorrect
2. Repair link missing on opposite end from shackle
3. Shackle pin nonfunctional
4. Chain incorrect type
5. Damage

Minor — AQL 4.0 percent defective
51. Shackle pin chain improperly assembled
52. Shackle incorrect type
53. Rust or contamination
54. Screw pin diameter incorrect

Incidental
None

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4.2.6.8 Leg - Assembly (less E28-12-13). - (Dwg. E28-12-11)

Categories and defects

Major — AQL 1.0 percent defective

1. Component missing
2. Leg length incorrect
3. Leg outside diameter incorrect
4. Leg bent or deformed
5. Welding improper
6. Eye thickness through hole incorrect
7. Eye hole diameter incorrect
8. Eye hole location from radius incorrect
9. Eye length hole to shoulder incorrect

Minor — AQL 4.0 percent defective

51. Lugs and eye relative alignment incorrect
52. Lugs location incorrect
53. Eye cocked in leg
54. Lug inverted
55. Lug hole diameter incorrect
56. Damage or deformation

Incidental — AQL 10 percent defective

101. Lug countersink missing
102. Lug thickness incorrect

4.2.6.9 Hoisting unit, tripod, drum, M1-assembly (packed byt not sealed). - (see 5.1)

Categories and defects

Major — AQL 1.0 percent defective

1. Strapping, (tripod) missing or loose
2. Contents loose in box
3. Box damaged or incorrect
4. Marking missing or incorrect
5. Component missing
6. Damage or deformation

Minor — AQL 4.0 percent defective

51. Chain not assembled to eye
52. Hoist and sling improperly wrapped
53. Hoist and sling improperly packaged
*54. Sealing improper

Incidental

None

* When shipped as spare parts only.

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4.3 Functioning test. - The hoisting unit shall be tested as follows: Attach the hoisting unit to an overhead support which will withstand vertical downward loads of a minimum of 1 ton. Attach the other end or load hook of the hoist using sling to a dead weight of 1000 ± 50 pounds. When the hand chain is pulled enough so that the load is clear of the floor or ground attach a spring scale to the hand chain and transfer the pull to this scale This scale shall be calibrated to 150 pounds in 5 pound increments. The scale shall be read while a slow steady pull is exerted on the hand chain.

5. PREPARATION FOR DELIVERY

5.1 Preservation and packaging.

5.1.1 Level A.

5.1.1.1 Tripod. - The tripod shall be collapsed. The outer end of each chain shall be detached from the neighboring leg and extended along the leg to which it is permanently attached. The anchor shackle shall be secured to an eye located near the top of the leg. The tripod shall then be secured in the collapsed position with steel wire or strapping.

5.1.1.2 Differential hoist. - All exposed and operating surfaces of the differential hoist shall be protected by the application of medium preservative lubrication oil conforming to Specification MIL-L-3150. The hoist shall then be wrapped in grease-proof barrier material conforming to grade A of Specification MIL-B-121, and placed in a $V3_S$, type II, class 3, grade 3, style RSC fiberboard box conforming to Specification PPP-B-636. The seams, corners, and joints of the box shall be sealed with 3-inch wide pressure-sensitive adhesive tape conforming to type I or II, class I of Specification PPP-T-60.

5.1.1.3 Hoist sling. - The sling shall be placed in a $V3_S$, type II, class 3, grade 3, style RSC fiberboard box conforming to Specification PPP-B-636. The seams, corners and joints of the box shall be sealed with 3-inch wide pressure-sensitive adhesive tape conforming to type I or II, class I of Specification PPP-T-60.

5.1.2 Level C.

5.1.2.1 Tripod. - The tripod shall be collapsed. The outer end of each chain shall be detached from the neighboring leg and extended along the leg to which it is permanently attached. The anchor shackle shall be secured to an eye located near the top of the leg. The tripod shall then be secured in the collapsed position with steel wire or strapping.

5.1.2.2 Differential hoist. - The hoist shall be placed in a type II, class 1, style RSC fiberboard box conforming to Specification PPP-B-636. The box shall be closed in accordance with the applicable requirements thereof.

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5.1.2.3 Hoist sling. - The sling shall be placed in a type II, class 1, style RSC fiberboard box conforming to Specification PPP-B-636. The box shall be closed in accordance with the applicable requirements thereof.

5.2 Packing

5.2.1 Level A. - A tripod, differential hoist, and a hoist sling, packaged as specified in 5.1.1, shall be packed in a nailed wood box constructed for a difficult load in accordance with class 2, style 2 or 4 of Specification PPP-B-621 in such a manner that a tight pack is obtained. The packaged hoist shall be placed in a compartment made in one end of the box. The approximate inside dimensions of the box shall be 11 1/2 feet long by 10 1/2 inches wide by 9 inches deep. The box shall be closed and strapped on accordance with the appendix to Specification PPP-B-621.

3.2.2 Level B. - A tripod, differential hoist, and a hoist sling, packaged as specified in 5.1.1, shall be packed in a nailed wood box constructed for a difficult load in accordance with class 1, style 2 or 4 of Specification PPP-B-621 in such a manner that a tight pack is obtained. The approximate inside dimensions of the box shall be 11 1/2 feet long by 10 1/2 inches wide by 9 inches deep. The box shall be closed in accordance with Specification PPP-B-621 and strapped in accordance with the appendix thereof.

5.2.3 Level C. - A tripod, differential hoist, and a hoist sling, packaged as specified in 5.1.2 shall be packed in a nailed wood box conforming to class 1, style 2 or 4 of Specification PPP-B-621 in such a manner that a tight pack is obtained. The packaged hoist shall be placed in a compartment made in one end of the box. The approximate inside dimensions of the box shall be 11 1/2 feet long by 10 1/2 inches wide by 9 inches deep. The box shall be closed in accordance with Specification PPP-B-621 and strapped in accordance with the appendix thereof.

5.3 Marking. - In addition to any special marking required by the contract or order, all interior and exterior shipping containers shall be marked in accordance with Standard MIL-STD-129.

6. NOTES

6.1 Ordering data. - Procurement should specify the following:

a. Title, number and date of this specification
b. Selection of applicable levels of preservation, packaging and packing

6.2 Stock number. - The Federal Stock number for this item is 3950-368-6182.

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